

Nimmagadda Laasya

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Professional Summary

Final-year Computer Science graduate seeking an **AI / Machine Learning Intern** position to contribute to applied AI and production-oriented ML workflows. Skilled in end-to-end pipelines covering data preprocessing, exploratory data analysis, supervised and unsupervised model development, evaluation, and API-based deployment. Published at IEEE ICCCNT 2025 (IIT Indore) and AIHC 2025 (NIT Calicut). Proficient in Python, Scikit-learn, NumPy, Pandas, NLP (DistilBERT), and computer vision (OpenCV), with hands-on exposure to XAI frameworks and cloud platforms (AWS, Azure).

Publications

A Hybrid ARIMA-LSTM Framework for LAN Latency Prediction with Stability Classification and Anomaly Detection

Accepted and Presented – 16th IEEE ICCCNT 2025, IIT Indore

- Developed a hybrid time-series model combining ARIMA (linear) and LSTM (nonlinear) for LAN latency prediction with anomaly detection and stability classification.

An Explainable Counterfactual ML Framework for Personalized Obesity Risk Prediction and Lifestyle Optimization

Accepted and Presented – AIHC 2025, NIT Calicut

- Built an XAI framework using Random Forest and DICE counterfactual explanations, achieving **86.95% accuracy** with personalized lifestyle recommendations for preventive healthcare analytics.

Professional Experience

AI/ML Intern – Edunet Foundation

Jan 2023 – Mar 2023

- Applied supervised and unsupervised ML algorithms to real-world structured datasets in an Agile/Scrum environment, improving model accuracy by **12%** through data preprocessing, feature engineering, and hyperparameter tuning.
- Conducted EDA to identify patterns and outliers; prepared train/validation/test splits and documented experimental workflows and results for team review.

Projects

LLM-Assisted Insider Threat Detection via Behavioral NLP | *Python, DistilBERT, Isolation Forest, Flask* May 2026

- Engineered a hybrid AI threat detection pipeline on the CERT Insider Threat Dataset (4,000+ records), integrating DistilBERT for NLP-based behavioral analysis and Isolation Forest for unsupervised anomaly detection across 3 data sources.
- Implemented multi-modal risk scoring with weighted score fusion, improving threat detection interpretability by **25%** and generating automated high-risk alerts via a Flask API and live dashboard.

Explainable ML Framework for Obesity Risk Prediction | *Python, Random Forest, DICE, Scikit-learn* Nov 2025

- Built a supervised ML framework on the UCI Obesity Dataset (**2,111 samples, 17 features**), achieving **86.95% cross-validation accuracy** across 6 risk categories with full preprocessing and model evaluation pipelines.
- Integrated XAI using DICE counterfactual explanations to generate interpretable, personalized health recommendations; translated ML outputs into actionable lifestyle insights for preventive healthcare analytics.

FIP Image Editor Pro | *Python, OpenCV, Tkinter, NumPy*

Sept 2025

- Developed a computer vision desktop application with 5+ OpenCV algorithms (CLAHE, bilateral filtering, noise reduction, edge detection) and a custom Tkinter GUI with real-time RGB histogram visualization per operation.
- Applied adaptive histogram equalization (CLAHE) across multiple color channels for contrast enhancement; designed interactive ROI selection tools enabling region-specific filter application.

Fetal Health Bioinformatics Classifier | *Python, Scikit-learn, Gradient Boosting, Seaborn*

Jan 2026

- Trained a supervised multi-class model on the UCI Fetal Health Dataset (**2,126 samples, 21 CTG features**) using Gradient Boosting and Random Forest with StandardScaler preprocessing and stratified cross-validation.
- Evaluated performance via ROC/AUC, confusion matrices, and feature importance across 3 diagnostic categories (Normal, Suspect, Pathological); documented algorithm comparisons and experimental findings.

Education

SRM University AP (Amaravati): B.Sc. in Computer Science | CGPA: 8.0

2022 – 2026

Certifications

Microsoft Learn: Azure ML Workspace & Responsible AI Badge

Jan 2026

AWS Academy: Cloud Foundations Training
Jan 2026

Skills

Programming: Python, Java, SQL, C, C++ **Libraries & Tools:** Scikit-learn, Pandas, NumPy, OpenCV, Flask, Tkinter, NetworkX, MySQL

AI / ML: Supervised Learning, Unsupervised Learning, NLP, Computer Vision, Data Preprocessing, Model Evaluation, EDA, Feature Engineering, Random Forest, Gradient Boosting, Isolation Forest, DistilBERT, XAI

Cloud & Core CS: AWS, Azure ML, Data Structures & Algorithms, Statistics, Linear Algebra